

AéroBox

A configurable transport-drone concept

Jean Tixeront and Matis Lebrun



Original project render. Academic concept, not a flight-tested aircraft.

The idea

AéroBox explores how a compact drone could adapt to different small-payload delivery missions. The project connects a parametric CATIA assembly with a configuration matrix, a bill of materials and a proposed manufacturing strategy.

The design takes inspiration from delivery-oriented drones and compact, agile aircraft. Amazon Prime Air and Red Bull are references only: neither company is a partner or an endorser of this academic project.

16 design variants

2 material families

Concept-stage CAD deliverable. No measured flight-performance claim.

From a mission to an assembly

01 / Requirements

The proposed product transports a light payload and is intended for vertical take-off and landing. Payload protection, access to electronics, stability and maintenance guide the layout. These are intended functions, not demonstrated operational capabilities.

02 / Parametric configuration

The design matrix combines arm lengths of 0.30 and 0.50 m, propellers with two to five blades, and carbon-fibre or glass-fibre material options. The resulting 16 variants structure comparison within a common assembly. Mass and flight performance still need to be calculated and validated.

03 / Product architecture

The central body groups the payload and electronic integration. Arms locate the motors, while propellers, protective elements and landing supports complete the layout. CATIA parameters make it possible to explore geometric changes without starting a new model for each variant.

04 / Manufacturing and sourcing

The proposed process depends on the part family: laser cutting or CNC machining for plates, tube cutting and drilling for arms, printing or machining for connectors, and composite or printed prototypes for fairings. Electrical components are treated as bought-in items requiring mechanical integration.

Supplier scope matters

The subcontracted contribution described in the source report is limited to propeller drawings. It does not establish manufacture, dynamic balancing or flight validation of the propellers or complete drone.

What the project demonstrates

Engineering deliverables

A parametric CAD assembly, project renders, a configuration matrix and documentation of components, manufacturing processes and supplier responsibilities.

Business relevance

The work links a user need to product choices. It provides a basis for discussing variant complexity, make-or-buy decisions, maintenance and the evidence needed before presenting a concept as a usable product.

Not yet validated

Final mass and centre of gravity, material assignments, interference checks, motor and battery selection, endurance, payload capability and flight behaviour. Proposed mission speeds or distances are not measured results.

Before any flight testing

Complete the assembly and clearance checks, verify the configuration changes, confirm suitable materials and motor alignment, and use appropriately validated and balanced propellers. A render alone cannot establish structural or flight safety.

Sources and attribution

Based on the original AéroBox presentation (May 2026), the General Concept of the Hybrid Transport Drone overview, and the Subcontracting Specification, BOM Overview and Manufacturing Strategy report (15 May 2026). Project co-developed by Jean Tixeront and Matis Lebrun. Portfolio brief prepared in September 2026.

Full project and original manufacturing report

tixe1996.github.io/juan-tixeront/projects/aerobox/